YieldFlo User Manual

YieldFlo is a yield monitoring application that works alongside AgOpenGPS (AOG). It records yield rate, moisture, and GPS position continuously while harvesting, and produces spatial yield maps and job reports.

YieldFlo receives GPS and section control data from AOG over a local network (UDP). A YieldFlo hardware module connects to the combine's clean-grain elevator and moisture sensor and sends live readings to the PC application.

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1. Getting Started

Requirements

Item	Requirement
Operating system	Windows 10 or Windows 11
.NET Framework	4.8 (pre-installed on Windows 10 May 2019 Update and later)
AgOpenGPS	Running on the same PC or local network

No installer is required. Extract the YieldFlo folder to any location and run `YieldFlo.exe`.

Hardware module

- YieldFlo module (based on ESP32) mounted on the combine
- Connected to the clean-grain elevator optical sensor
- Connected to the capacitance moisture sensor (optional)
- Communicates with the PC via WiFi or CAN bus

Network

- **WiFi mode:** connect the PC to the same WiFi network as the YieldFlo module, or directly to the module's access point
- AOG must be running and broadcasting GPS/section data on the local network

Recommended first-run setup

On first launch YieldFlo creates a default crop, header, and profile. Before starting your first job:

1. Open **Menu** → **Settings** and select your preferred units (Imperial or Metric)
2. Open **Menu** → **Profiles** and enter your combine ID
3. Open **Menu** → **Crops** and add or edit crops for your operation
4. Open **Menu** → **Headers** and enter the correct cutting width
5. Open **Menu** → **Fields** and add field names if desired

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2. Main Screen

The main screen displays live harvest data and job status.

Toolbar buttons

Button	Action
Menu	Open the main menu
Start	Start a new job (opens Jobs menu) or resume the paused active job
Pause	Manually pause recording
Stop	Stop and close the current job (requires confirmation)
Exit	Close YieldFlo
—	Switch to compact (mini) view — see below

Data panels

Panel	Description
YIELD	Instantaneous yield rate (bu/ac or t/ha), smoothed over recent readings
MOISTURE	Live grain moisture percentage from the capacitance sensor
SPEED	Ground speed received from AOG GPS
Totals area	Total area harvested, total mass, and average yield for the current job

Sensor bars

Two horizontal bar gauges below the data panels:

- **Elev Flow:** Grain obstruction percentage in the clean-grain elevator. Higher = more grain flowing.
- **Moisture:** Relative moisture sensor reading scaled to a 0–30% range.

Status bar

Indicator	Green	Orange	Red / Silver
GPS	AOG connected and sending position	—	No AOG data
Module	Module data received within 5 s	—	No module data
Comm	—	—	Shows active comm type (UDP or CAN)
Job name	Job recording (active)	Job paused	No active job (silver)

When YieldFlo displays a notification (e.g. export complete, error), a full-width message overlays the status bar for 10 seconds then clears automatically. Yellow text = informational; red text = error.

Compact (mini) view

Press **—** in the top-right corner of the main screen to switch to a small floating overlay that shows only the live yield reading. This is useful when running YieldFlo alongside AgOpenGPS on a single monitor.

The compact overlay can be dragged anywhere on screen. Press the restore button on the overlay to return to the full main screen. Its position is remembered between sessions.

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3. Jobs

Menu: Menu → Jobs

A job records yield, moisture, GPS position, and area for a single harvest session. Each job is linked to a crop, header, and profile.

Starting a job

1. Press **Menu** on the main screen, then **Jobs**
2. Press **New** and enter a job name
3. Select a **Crop**, **Header**, **Profile**, and optionally a **Field**
4. Enter any **Notes** for the job (optional)
5. Press **Start Job**

Recording begins automatically when harvesting conditions are met (speed > 0.5 km/h and AOG sections are on).

Managing jobs

The jobs list shows all saved jobs with their status, date, area, and linked field. Click a column header to sort by that column.

Action	Description
Load	Make a saved job the active job (prompts if another job is already active)
Delete	Permanently remove a job and all its recorded data — cannot be undone

Only one job can be active at a time.

Auto-pause

The job automatically pauses when harvesting stops:

- Machine speed falls below 0.5 km/h, **or**
- AOG turns all header sections off (e.g. turning on headland, travelling over harvested ground)

When auto-paused the job name turns orange in the status bar. Recording resumes automatically when harvesting conditions return.

Manual pause and resume

Press **Pause** on the main screen to manually pause recording. Press **Start** to resume.

Stopping a job

Press **Stop** → confirm. The job is saved and closed. All totals are written to the database. The job remains in the list and can be viewed in the Job Report.

Resume on start

If **Resume Job on Start** is enabled in Settings, the active job is automatically reloaded when YieldFlo restarts. Useful if the PC is rebooted during a harvest day.

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4. Crops

Menu: Menu → Crops

Crops store the grain-specific parameters used in yield calculations and reporting.

Crop settings

Field	Description
Name	Crop name (e.g. Wheat, Corn, Canola)
Category	Grain type category
Test Weight	Standard bushel weight (lbs/bu) — used to convert mass to bushels
Market Moisture	Standard moisture for yield reporting (e.g. 14.5% for wheat)

Adding a crop

1. Press **New** and enter a crop name
2. Set the test weight and market moisture
3. Press **Save**

Select a crop from the list to edit it. At least one crop must exist at all times.

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5. Headers

Menu: Menu → Headers

Headers define the cutting width of the front attachment. Width is used to calculate the area harvested per data point and to draw swath polygons on the yield map.

Header settings

Field	Description
Name	Header name (e.g. 30ft Draper, 8-row Corn Head)
Type	Header type category
Width	Cutting width in feet (Imperial) or metres (Metric)

Adding a header

1. Press **New** and enter a header name
2. Enter the cutting width
3. Press **Save**

At least one header must exist at all times.

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6. Profiles

Menu: Menu → Profiles

Profiles store combine-specific and calibration settings. Use a separate profile for each combine if running YieldFlo on multiple machines.

Profile settings

Field	Description
Name	Profile name (e.g. JD 9600, Case 2388)
Combine ID	Identifier for the module — must match the module configuration
Moisture offset	Fixed offset applied to all moisture readings for this profile
Moisture scale	%/count scale factor for the moisture sensor
Temperature offset	Temperature offset (°C)
Temperature scale	°C/count scale factor for the temperature sensor

> **Note:** Moisture and temperature calibration values are set in **Menu** → **Moisture Cal** and saved automatically to the active profile.

Adding a profile

1. Press **New** and enter a profile name
2. Enter the Combine ID to match your module
3. Press **Save**

At least one profile must exist at all times.

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7. Fields

Menu: Menu → Fields

Fields are optional location labels that can be assigned to jobs for organisation and reporting.

Adding a field

1. Press **New** and enter a field name
2. Press **Save**

Fields are not required. A job can be created without a field selected.

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8. Yield Calibration

Menu: Menu → Yield Cal

Yield calibration corrects the elevator sensor reading to match the actual mass of grain harvested. Two parameters are set here:

- **Sensor Baseline** — the obstruction reading with paddles running but no grain (set once at installation)
- **Yield Factor** — a multiplier that scales the sensor reading to match a weighed reference

Setting the baseline

1. Start the clean-grain elevator with no grain
2. Open **Menu** → **Yield Cal**
3. When the reading is stable, press **Set Baseline**
4. Confirm the prompt

> **Tip:** Run the empty elevator for at least 10 seconds before setting the baseline so the reading stabilises.

Processing delay

The processing delay accounts for the travel time of grain from the header to the elevator sensor. Set this value (in seconds) to match your combine. Typical range is 4–12 seconds.

Running a calibration pass

A calibration pass measures the actual mass of grain harvested during a known run, then adjusts the Yield Factor to match.

Preparation: Position a weigh wagon or grain cart to catch grain from the unloading auger and record start and end weights.

1. Open **Menu** → **Yield Cal**
2. Press **Start Run** — the app begins accumulating sensor data
3. Harvest a suitable area (at least one full tank is recommended)
4. Press **Stop Run**
5. Weigh the harvested grain
6. Enter the actual weight in the **Actual weight** field
7. Press **Apply Cal**

YieldFlo calculates a new Yield Factor and displays the result. Press **Save & Apply** to save the new factor to the active profile.

Manual factor adjustment

The Yield Factor can also be adjusted manually in the **Yield Factor** field. Increase to raise yield readings; decrease to lower them.

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9. Moisture Calibration

Menu: Menu → Moisture Cal

Moisture and temperature readings are both calculated the same way:

```
Value = raw_count × scale  +  offset
```

Both support single-point calibration. With offset set to 0, adjust scale until the displayed reading matches a reference instrument:

```
scale = known_value / raw_count
```

For example: if the raw count is 420 and a certified meter reads 18.5%, set scale to 18.5 ÷ 420 ≈ 0.044. The offset field can then be used for fine trimming against a second reference point if needed.

Moisture calibration procedure

1. Take a grain sample and measure moisture with a certified grain moisture meter
2. Open **Menu** → **Moisture Cal**
3. With the module connected and grain flowing, observe the live raw count
4. Set offset to 0 and adjust **%/count** scale until the displayed reading matches the meter
5. Press **Save**

Moisture calibration fields

Field	Saved to	Description
%	Active crop	Moisture offset — set to 0 for single-point scale calibration
%/count	Active profile	Scale factor converting raw ADS1115 counts to moisture percent

Temperature calibration

Same principle as moisture. The default scale of 0.0125 assumes an LM35 sensor (10 mV/°C at PGA = 4.096 V). With offset at 0, adjust **°C/count** until the reading matches a thermometer.

Field	Saved to	Description
°C	Active profile	Temperature offset — set to 0 for single-point scale calibration
°C/count	Active profile	Scale factor converting raw ADS1115 counts to degrees Celsius

> **Note:** Temperature calibration is optional. Temperature does not appear directly on the main screen.

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10. Yield Map

Menu: Menu → Yield Map

The yield map displays a colour-coded spatial plot of yield rate across the harvested area. Each data point is drawn as a filled swath polygon sized to the header width and oriented to the GPS heading at that point.

Colour scale

Yield is mapped to a five-colour gradient:

Colour	Yield
Blue	Lowest
Cyan	Low
Green	Mid
Yellow	High
Red	Highest

The legend at the bottom of the full-screen map shows the low and high yield values for the current data set.

Mini mode

When opened from the menu, the yield map starts as a small floating 300×300 overlay (mini mode). The title bar shows the active job name and zoom controls.

Control	Action
— / **+**	Zoom out / in
×	Close the map overlay
Click the map	Expand to full-screen mode
Drag the title bar	Move the overlay

The mini map position is remembered between sessions.

Full-screen mode

Click the map or press **×** to expand to full screen. The toolbar at the top provides:

Control	Action
Job selector (drop-down)	Choose which job to display
— / **+**	Zoom out / in
Print	Export the map as a PNG image — see below
Close	Return to mini mode

The map can be dragged in both mini and full-screen modes.

Live update during harvest

When the yield map is open and a job is actively recording, the map updates automatically every 5 seconds. New swath polygons are added without re-centering the view.

Exporting the map as a PNG

Press **Print** in the full-screen toolbar to export the current map view (including the legend) as a PNG image.

1. A save dialog opens with a default filename of `JobName_Date.png`
2. Choose a folder and confirm
3. The image is saved and a confirmation appears in the status bar

The export folder is remembered for subsequent exports.

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11. Job Report

Menu: Menu → Reports

The job report shows a summary of a completed or active job. Select a job from the list on the left to view its details.

Report contents

Field	Description
Job name	Name entered when the job was created
Field	Field name assigned to the job (if set)
Area	Total area harvested (ac or ha)
Total	Total mass harvested (bu or tonnes)
Avg Yield	Average yield rate across all data points
Avg Moisture	Average grain moisture across all data points
Data Points	Number of recorded GPS data points

Printing a report

Press **Print** to send the job summary to the system print dialog. The report is formatted as plain text suitable for any printer.

CSV export

Press **Export CSV** to save all raw data points for the selected job.

1. A save dialog opens with a default filename of `JobName_Date.csv`
2. Choose a folder and confirm
3. The file is saved and a confirmation appears in the status bar

The export folder is remembered for subsequent exports. The CSV format is compatible with the AgOpenGPS Rate Controller yield overlay.

Column	Description
Timestamp	Date and time of the reading
Latitude	GPS latitude (decimal degrees)
Longitude	GPS longitude (decimal degrees)
WidthMeters	Header cutting width (metres)
Yield_kgha	Yield rate in kg/ha
ElevationMeters	GPS elevation (metres)
Speed_kmh	Ground speed (km/h)
Heading	GPS heading (degrees)
Moisture_pct	Grain moisture (%)
HaAccumulated	Cumulative area at this point (ha)
Sensor1Raw	Raw elevator sensor ratio

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12. Settings

Menu: Menu → Settings

Units

Setting	Options
Units	Imperial (bu/ac, mph, acres, lbs) or Metric (t/ha, km/h, ha, tonnes)

Changing units takes effect immediately. Historical data is stored in metric units internally and converted for display.

Theme

Setting	Options
Theme	Dark or Light — changes the main screen background and gauge colours

Module communication

Setting	Description
WiFi (UDP)	Module communicates over UDP WiFi on port 30100. Connect the PC to the module's WiFi access point or a shared network.
CAN	Module communicates over CAN bus via a USB CAN interface.
CAN Driver	Select the CAN interface driver (SLCAN, InnoMaker, or PCAN).
COM Port	Serial/USB port for the CAN interface (CAN mode only).

Resume Job on Start

When enabled, YieldFlo automatically reloads the active job when the app starts. Useful if the PC is rebooted during a harvest day.

Saving settings

Press **Save & Apply** to save and apply all settings immediately.

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13. Language

Menu: Menu → Language

YieldFlo supports the following languages:

Code	Language
en	English
de	German
fr	French
hu	Hungarian
nl	Dutch
pl	Polish
ru	Russian
lt	Lithuanian

Changing language

1. Press **Menu → Language**
2. Select the desired language from the list
3. Press **Save & Restart**

YieldFlo will restart automatically with the new language applied.

> **Note:** If AgOpenGPS is already installed, YieldFlo will automatically use the same language on first launch.

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14. Troubleshooting

GPS indicator stays red

- Confirm AgOpenGPS is running on the same PC or local network
- Check that both apps are on the same subnet
- Verify the network configuration in YieldFlo Settings

Module indicator stays off

- Check the module is powered and the WiFi or CAN connection is active
- In WiFi mode, confirm the PC is connected to the correct network
- Check the module LED for status indication
- Restart the module and wait 10–15 seconds for it to connect

Yield reads zero or very low

- Check the elevator optical sensor alignment and wiring
- Verify the Sensor Baseline is set correctly (set with elevator running empty)
- Confirm the processing delay is appropriate for your combine

Moisture reads incorrectly

- Verify the moisture sensor is installed and wired correctly
- Check that the moisture calibration (offset and scale) is set in the active profile via **Menu → Moisture Cal**
- Compare against a certified grain moisture meter and adjust the calibration

App crashes on start

- Check `YieldFlo_Crash.log` in the application folder for details

Job does not resume after restart

- Ensure **Resume Job on Start** is enabled in Settings
- The job must have been active (not stopped) when YieldFlo was closed

AOG UDP failed to start

- If AgOpenGPS Rate Controller (RC) is running, both RC and YieldFlo must be built from current source for UDP port sharing to work
- YieldFlo will still function for module data — only GPS reception will be affected

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YieldFlo is designed for use with AgOpenGPS. It is not a certified weighing system and should not be used as the *sole basis for grain settlement.*